

ii. For all $t > 0$, we have $\tilde{F}(tX, t^\beta Y) = t^{d-e} \tilde{F}(X, Y)$ and $\tilde{G}(tX, t^\beta Y) = t^{d-e} \tilde{G}(X, Y)$, where $e = e_F = e_G$.

Proof. Let $\beta = r/s$, where $r > s > 0$ and $\gcd(r, s) = 1$. Since F and G are β -quasihomogeneous polynomials of degree d , we have:

$$F(X, Y) = \sum_{k=0}^m a_k X^{d-rk} Y^{sk} \quad \text{and} \quad G(X, Y) = \sum_{k=0}^n b_k X^{d-rk} Y^{sk},$$

where $a_m \neq 0$, $b_n \neq 0$, and $0 \leq m, n \leq \lfloor d/r \rfloor$. From these equations, it follows that $e_F = d - rm$ and $e_G = d - rn$. Thus, in order to prove that $e_F = e_G$, it suffices to show that $m = n$.

Let us prove that $m = n$. Since $f_+(t) = \sum_{k=0}^m a_k t^{sk}$ and $f_-(t) = \sum_{k=0}^m (-1)^{d-rk} a_k t^{sk}$, we have $\deg f_+ = \deg f_- = sm$; and since $g_+(t) = \sum_{k=0}^n b_k t^{sk}$ and $g_-(t) = \sum_{k=0}^n (-1)^{d-rk} b_k t^{sk}$, we have $\deg g_+ = \deg g_- = sn$. Also, by hypothesis, f_+ and f_- are Lipschitz equivalent to g_+ and g_- in some order. Hence, by Lemma 2.2, it follows that $sm = sn$. Therefore, $m = n$.

From now on, we drop the subscript and denote simply by e the multiplicity of X as a factor of either F or G . Let us prove the second part of the proposition. Since F is a β -quasihomogeneous polynomial of degree d , we have:

$$F(tX, t^\beta Y) = t^d F(X, Y) = t^d X^e \tilde{F}(X, Y).$$

On the other hand, since the multiplicity of X as a factor of F is equal to e , we have:

$$F(tX, t^\beta Y) = (tX)^e \tilde{F}(tX, t^\beta Y) = t^e X^e \tilde{F}(tX, t^\beta Y).$$

From these equalities, it follows that

$$\tilde{F}(tX, t^\beta Y) = t^{d-e} \tilde{F}(X, Y).$$

Clearly, the deduction above with F replaced with G yields the other identity. ■

Corollary 3.5. *Let $F, G \in \mathbb{R}[X, Y]$ be β -quasihomogeneous polynomials of degree d . If the height functions of F and G can be arranged into pairs of Lipschitz equivalent functions, then either both F and G are of the form cX^d , or neither of them is.*

Proof. Denote by e_F the multiplicity of X as a factor of F , and by e_G the multiplicity of X as a factor of G . By Proposition 3.10, F is of the form cX^d if and only if $e_F = d$, and G is of the form cX^d if and only if $e_G = d$. Suppose that the height functions of F and G can be arranged into pairs of Lipschitz equivalent functions. Then, by Proposition 3.11, $e_F = e_G$, so that $e_F = d$ if and only if $e_G = d$. Therefore F is of the form cX^d if and only if G is of the form cX^d . ■

The next proposition shows how to determine whether any two polynomials of the form cX^d , with $c \neq 0$ and $d \geq 1$, are $\mathcal{R}$ -semialgebraically Lipschitz equivalent.

Proposition 3.12. *Let $F(X, Y) = aX^d$ and $G(X, Y) = bX^d$, where $a, b \in \mathbb{R} \setminus \{0\}$ and $d \geq 1$.*

- i. *If d is even, then F and G are $\mathcal{R}$ -semialgebraically Lipschitz equivalent if and only if a and b have the same sign.*
- ii. *If d is odd, then F and G are $\mathcal{R}$ -semialgebraically Lipschitz equivalent.*